

Neuro-Symbolic ASR for Dysarthric Speech

via Articulatory-Constrained Posterior Reshaping

Helping speech recognition systems better understand people with motor speech disorders.

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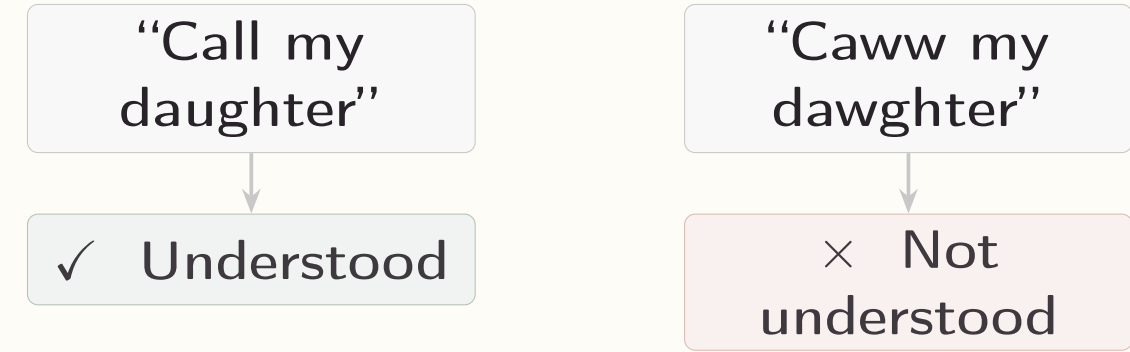
We do not treat dysarthric speech as **noise**.
We model it as **structured** articulatory variation.

15 Speakers | 16,552 Utterances | 95M Parameters | 47 ARPABET Tokens

Why Speech Matters

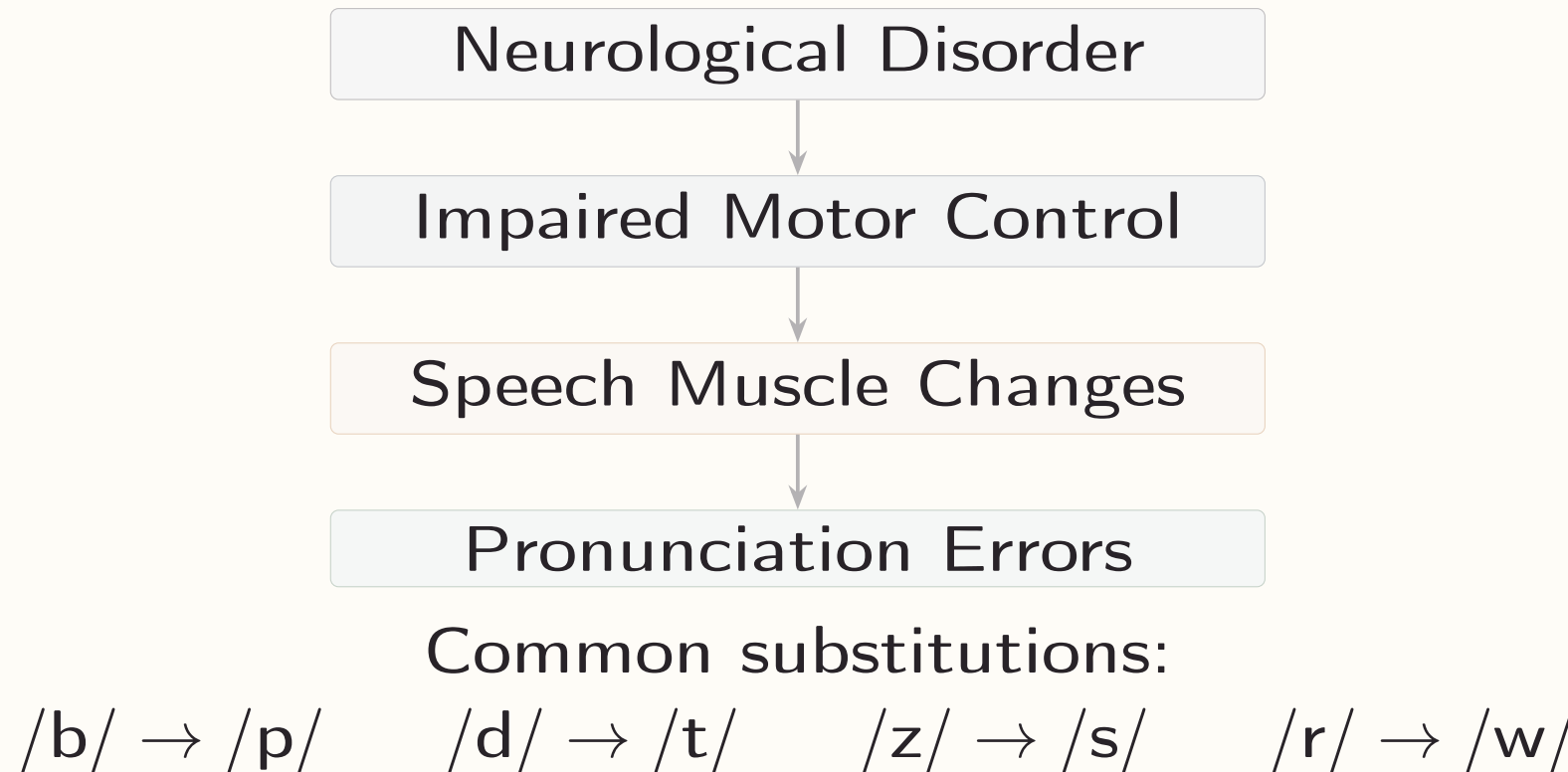
Voice assistants are everywhere, but millions with motor speech disorders cannot use them. The barrier is not language, but impaired speech production.

Typical Speech Dysarthric Speech



What Is Dysarthria?

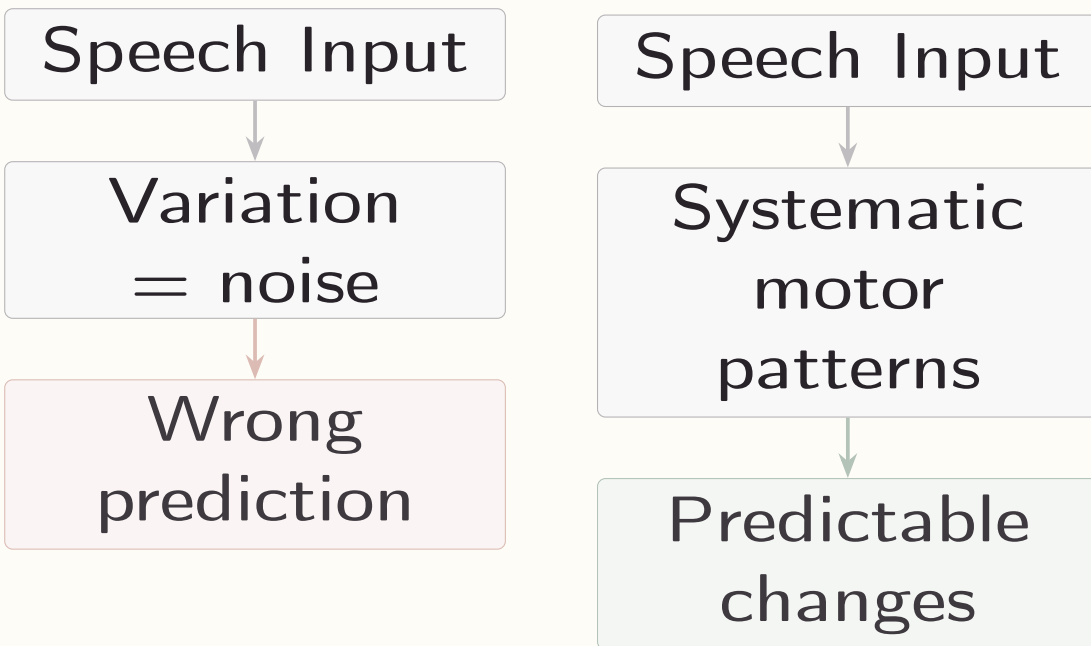
Dysarthria is a motor speech disorder caused by neurological conditions such as stroke, ALS, or cerebral palsy. Impaired muscle control leads to predictable pronunciation errors.



Why Conventional ASR Fails

Conventional ASR treats every acoustic irregularity as noise. When /b/ sounds like /p/, it sees corruption, not a known pattern.

Standard ASR Clinical Reality

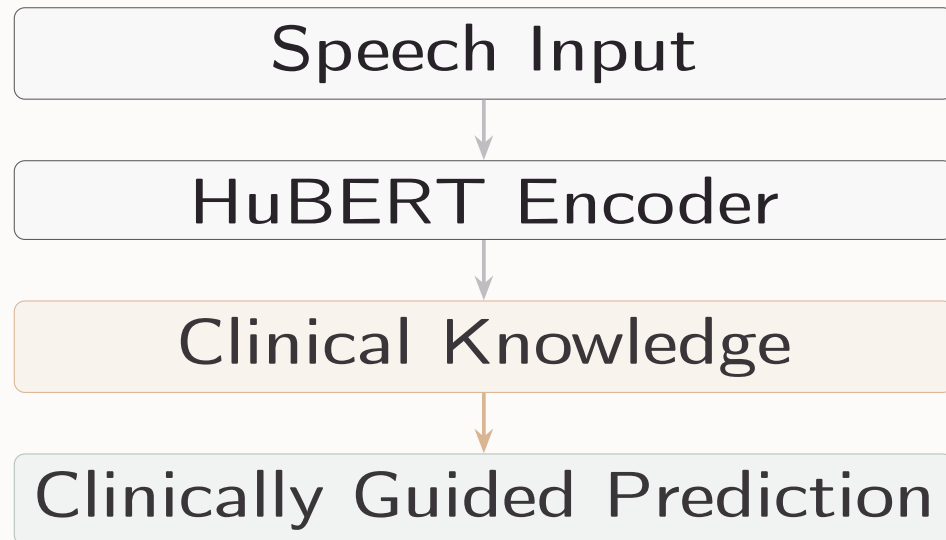


Dysarthric pronunciation errors are structured, not random.

This motivates incorporating explicit articulatory knowledge into ASR.

Key Insight

Clinical knowledge steers HuBERT toward articulatorily plausible outputs instead of treating dysarthric variation as acoustic noise.



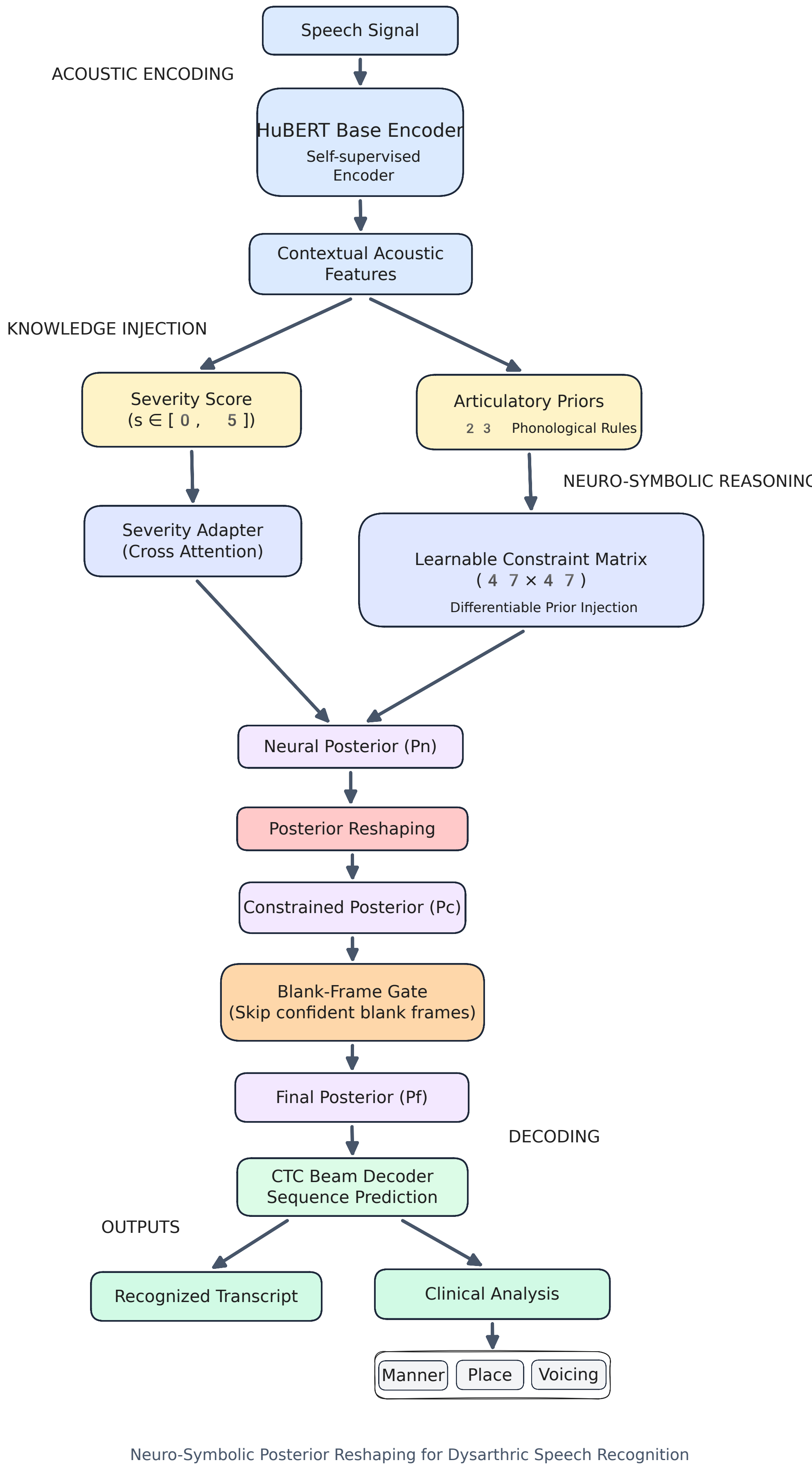
Our Contributions

- Learnable Constraint Matrix (LCM)
- Blank Frame Gate
- Posterior Reshaping without Retraining
- Improved Recognition Accuracy

From Problem to Solution



Proposed Framework



Neuro-Symbolic Posterior Reshaping for Dysarthric Speech Recognition

How the LCM Works

The Learnable Constraint Matrix (LCM) is a learnable 47 × 47 matrix initialized from clinical knowledge of three articulatory dimensions:

- Manner**: How airflow is shaped
- Place**: Where constriction occurs
- Voicing**: Whether vocal cords vibrate

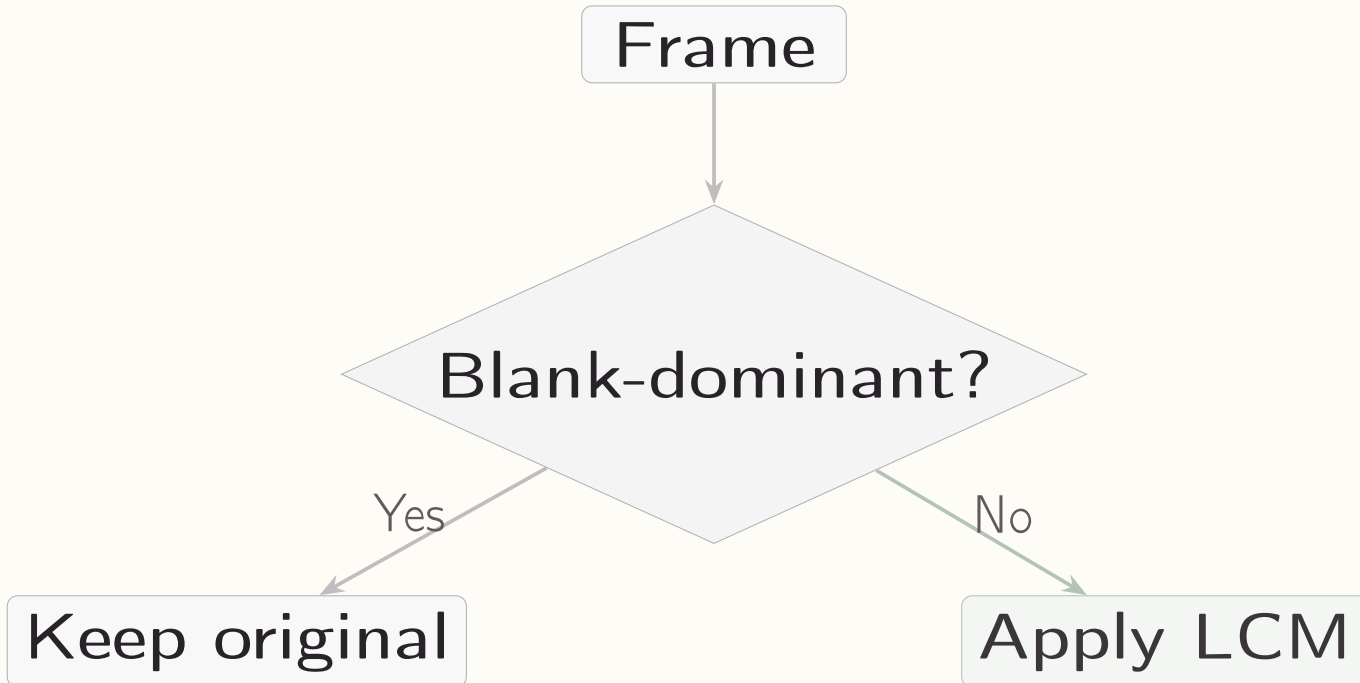
These constraints suppress implausible substitutions. When HuBERT is unsure between /b/ and /p/, the LCM reweights probabilities based on clinical plausibility:

Before LCM After LCM
/b/: 0.35 /p/: 0.30 /b/: 0.72 /p/: 0.18
/d/: 0.20 /t/: 0.15 → /d/: 0.06 /t/: 0.04
HuBERT posterior: unsure LCM-guided reshaping

This mechanism enables clinically guided decoding without retraining the acoustic model.

Blank Frame Gate

The Blank Frame Gate decides per frame whether to apply the LCM or trust the original prediction.



About 65% of frames are blank-dominant and pass through unchanged. The LCM is applied to the remaining 35%. The gate adds **zero additional parameters** and operates on phoneme classifier softmax posteriors. Under beam search: **helpful** 9.16%, **harmful** 3.78% (2.4:1 helpful-to-harmful ratio).

Experimental Setup

Dataset	Base Model	Validation
TORG0 dysarthric corpus	HuBERT Base	15-fold LOSO-CV
16,552 utterances	95M parameters	Bootstrap 95% CI
15 speakers	47 ARPABET tokens	

5.0% Relative PER Reduction

Over HuBERT + adapter | $p < 0.001$

We next evaluate whether these constraints improve recognition on real dysarthric speech.

Key Results

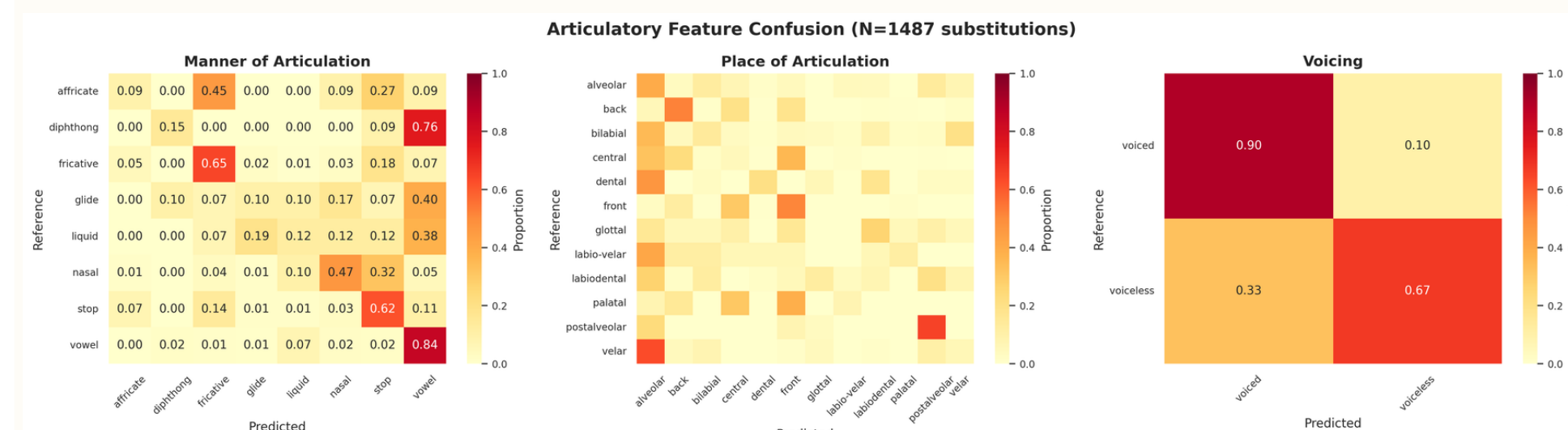
0.285	0.336	35%
Macro PER	Macro WER	Frames processed
80.5%	95.8%	90.4%
Manner accuracy	Voicing accuracy	Place accuracy

Qualitative Correction

Input	Baseline (HuBERT)	DysarthriaNSR (Ours)
"big dog"	"pig tog"	"big dog"
"red car"	"wed car"	"red car"
"this look"	"dis look"	"this look"

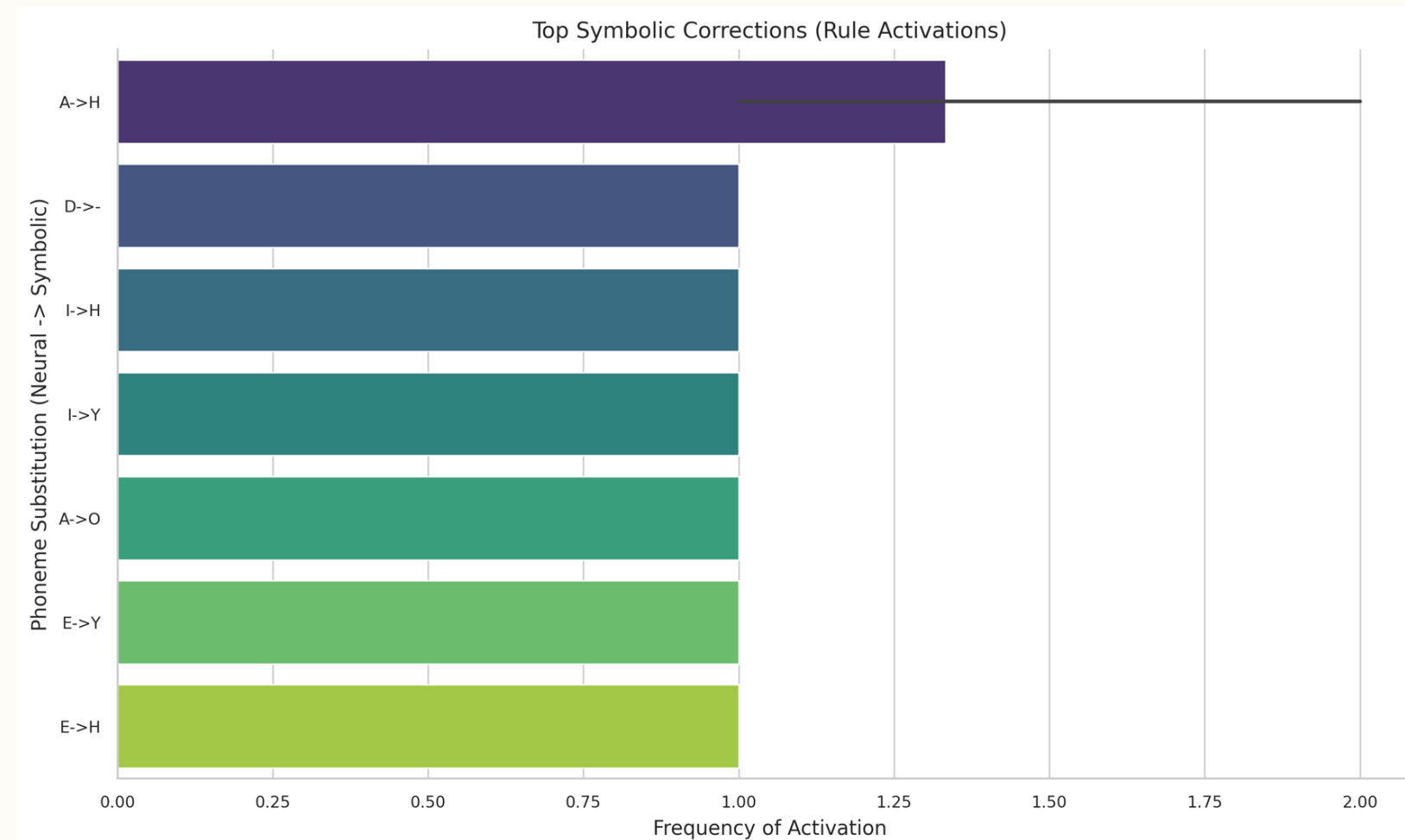
LCM corrects articulatory substitutions by recognizing the underlying clinical pattern.

Articulatory Confusion Analysis



Devoicing is the dominant confusion pattern (32%).

Rule Impact



LCM regularization consistently improves performance across all articulatory prediction heads.

Why This Matters

This work bridges speech technology and clinical care. Encoding articulatory knowledge directly into ASR enables personalized recognition for motor speech disorders, without large dysarthric datasets. Articulatory predictions provide interpretable diagnostic cues, making the system accurate and trustworthy.

Conclusion

Key Takeaway

Dysarthria is not **noise**, it is a pattern.

When ASR understands that pattern, recognition improves.

A neuro-symbolic framework that reshapes HuBERT's posterior via a differentiable confusion matrix initialized from clinical priors. On TORG0: **5.0%** relative PER reduction ($p < 0.001$), with the LCM applied to ~35% of frames.

Questions?



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